

# Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka

Project ID: 25-26J-423



# Our Team



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# Introduction to the Problem



Students in grades 7 to 9 in Sri Lanka rely on rote learning. They memorize instead of applying knowledge.



History is losing value in the curriculum. Students avoid it. This weakens cultural awareness and critical thinking.



Science and Math concepts are abstract. Students lack labs and tools to practice.



English is taught without real-life communication. Students cannot build confidence in speaking.

# Social or Research Impact of the Solution



The solution promotes active learning through VR. Students interact with Math, Science, English, and History in historical settings.



It raises motivation through gamification with points, missions, and rewards.



It supports bilingual delivery in Sinhala and English. This helps rural students.



It preserves cultural identity by embedding historical content.



It aligns with the national curriculum. Students gain practical support.



# Previous Research and Shortcomings



Global VR apps exist but are not syllabus aligned. They focus on general learners.



Local projects were prototypes. They lacked integration with other subjects and had no bilingual support.



Existing tools often need internet access, which rural schools lack.



This project builds a syllabus aligned VR platform. It integrates all four subjects inside historical contexts. It includes bilingual narration and offline use.

# Constraints and Limitations



Hardware access is limited to available Meta Quest devices.



Teacher training is required for classroom use.



Student readiness is uneven, especially in rural areas.



The project will begin with pilot testing in selected schools. Scaling will depend on results and resources.



# Learning History in Gamified VR Educational Platform

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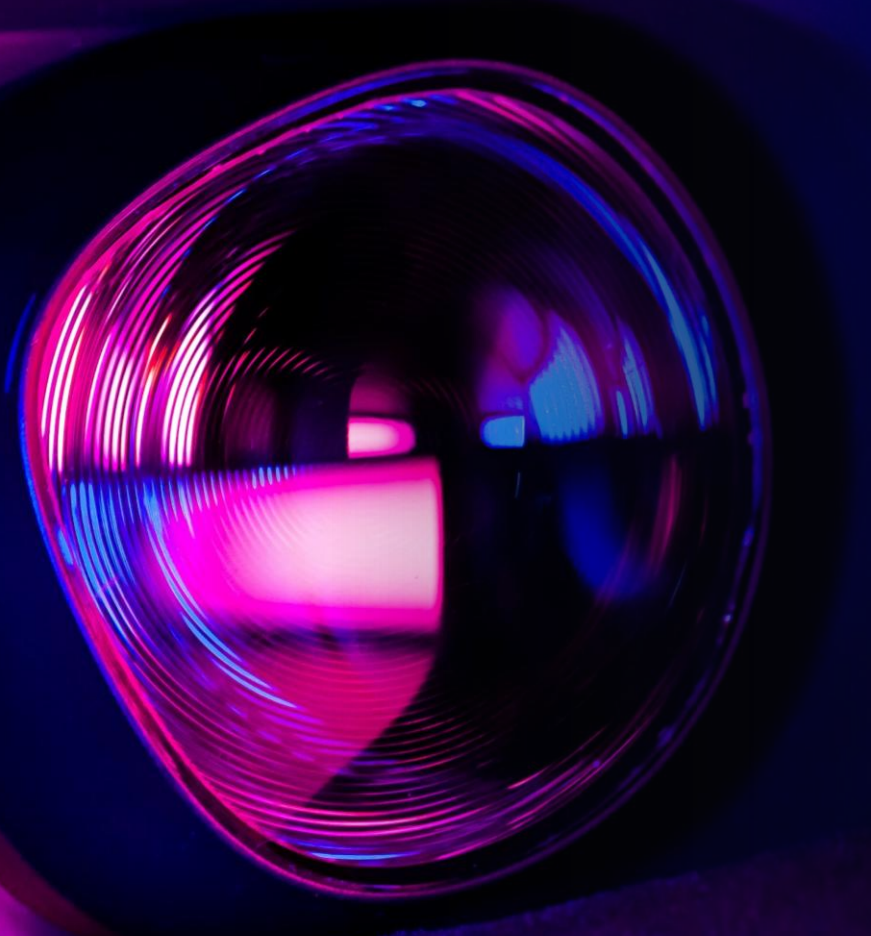
Jayalath K.M.K.D

Specialization: Interactive Media





# Proven Gap & Creative Solution



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Students memorize history facts without context, schools lack interactive tools especially in rural areas, existing VR history projects are only prototypes without syllabus alignment or gamification, and there is a risk of history being undervalued if it becomes an elective.

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Build VR historical worlds aligned to the Grade 7–9 syllabus, embed interactive missions with storytelling and gamification, integrate Math, Science, and English tasks within history environments, and provide bilingual Sinhala-English narration and subtitles for inclusivity.

# Research Gap

| Research Name                                           | Current Development (What they researched / where they stopped)   | Future & Special Features (Our Module Contribution)                                                    |
|---------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| VR in History Education: Systematic Quantitative Review | Emphasis on excitement/interest; limited measured learning gains. | Deliver measurable outcomes via Sinhala–English missions, assessments, and offline classroom packs.    |
| Challenges & Implications of VR in History Education    | Lists barriers (cost, UX, training); no practical solutions.      | Provide cost-aware offline VR, simple Sinhala–English interfaces, teacher training, and safety/ethics. |
| Past Lives, Present Learners (History in VR)            | Builds empathy; weak factual retention.                           | Blend empathy with knowledge checks, Sinhala–English narration, and curriculum-aligned quests.         |
| Design & Testing of a Historical VR Experience          | Single-event focus; no broad curriculum integration.              | Cover multiple eras with Sinhala–English content, cross-subject links, and offline delivery.           |
| Immersive VR for Presence and Empathy                   | Measures empathy only; lacks learning assessment/practice.        | Add assessments + analytics, Sinhala–English support, and gamified tasks tied to syllabus outcomes.    |

# Shortcomings of Existing Systems and the Proposed Solution

| Feature                                                                         | Google Expeditions (History VR) | TimeLooper (Historical VR) | ClassVR | Edirisinghe et al. (Sri Lanka VR Heritage Prototype) | Proposed VR History Module |
|---------------------------------------------------------------------------------|---------------------------------|----------------------------|---------|------------------------------------------------------|----------------------------|
| Syllabus Alignment (Sri Lankan Grades 7–9)                                      | ✗                               | ✗                          | ✗       | ✗                                                    | ✓                          |
| Cultural Relevance (Sri Lankan Contexts)                                        | ✗                               | ✗                          | ✗       | ✓                                                    | ✓                          |
| Bilingual Support (Sinhala + English)                                           | ✗                               | ✗                          | ✗       | ✗                                                    | ✓                          |
| Coverage of Historical Learning (Critical Thinking, Identity, Heritage, Events) | ✗                               | ✗                          | ✗       | ✗                                                    | ✓                          |
| Gamification & Motivation Design                                                | ✗                               | ✓                          | ✗       | ✗                                                    | ✓                          |
| Offline/Low-resource Accessibility                                              | ✗                               | ✗                          | ✗       | ✗                                                    | ✓                          |
| VR Optimization (Meta Quest 3 / Headsets)                                       | ✓                               | ✓                          | ✓       | ✗                                                    | ✓                          |



# Application of Specialization-based Knowledge

## Key Domains of Specialization-based Knowledge



- Narrative design and cultural immersion.
- Interactive media for experiential history learning.
- Gamified storytelling techniques.

## Specific Latest Technologies in that Domain



- Unity 3D XR Toolkit for VR navigation.
- Blender for accurate 3D reconstructions of Sri Lankan heritage.
- Bilingual voice and subtitles.
- Gamification with points, quests, and progression.





## How to Evaluate and Prove Success



- Student engagement measured through surveys and observation.
- Knowledge retention on history topics tested with quizzes.
- Teacher assessments of accuracy and syllabus match.
- Metrics on mission completion and replay rates.

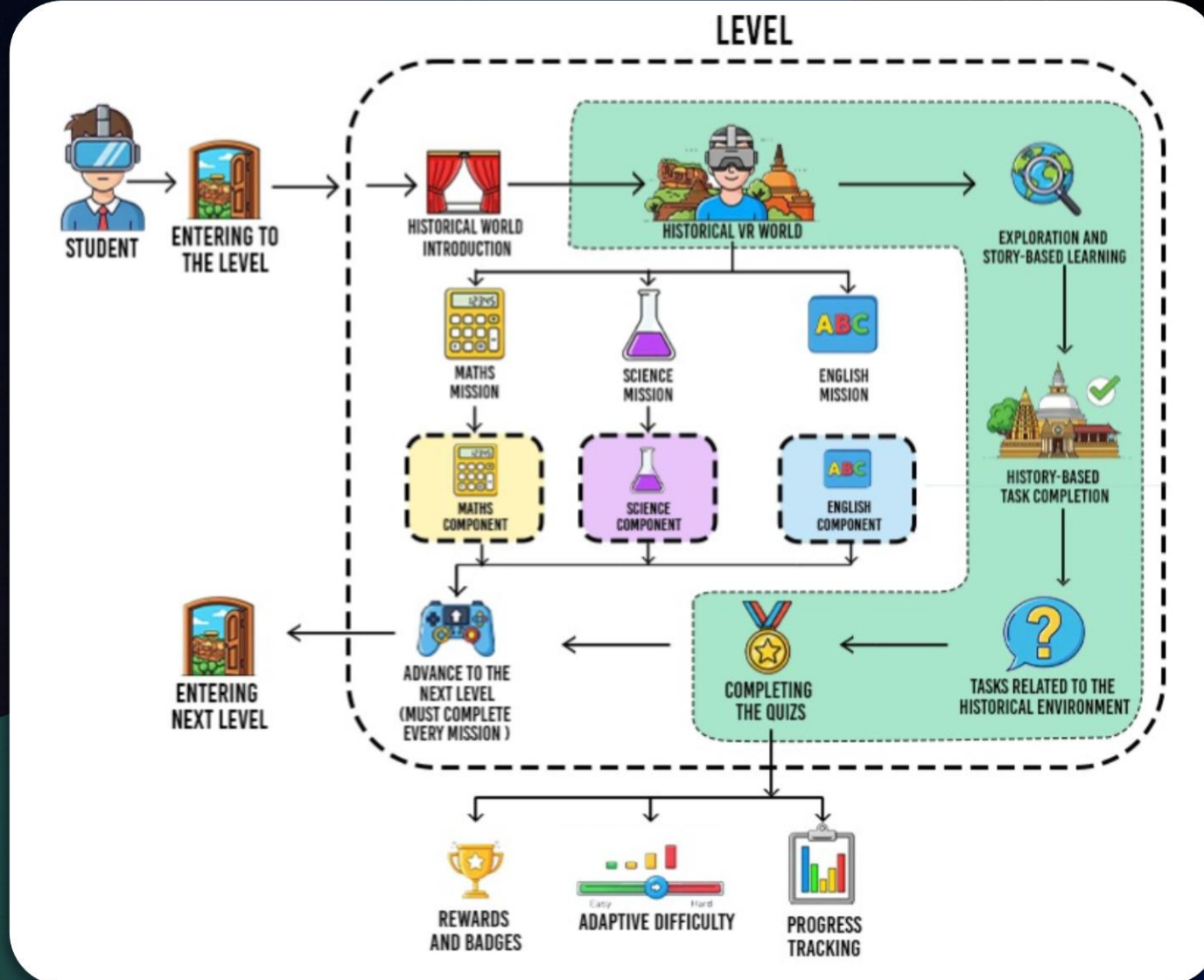
## Data Availability and Ethical Clearance



- National curriculum and NIE History guides.
- Surveys and interviews with students in two rural schools.
- School and parental consent for pilot testing.



# System Overview



# Real-world Scenario

- A Grade 8 student explores a VR scene of colonial Ceylon.
- They interact with historical characters and complete missions.
- Successful completion reveals timeline progression and cultural facts.
- Students use the module to connect history with other subjects in an integrated lesson.

# Functional Requirements

- Students can explore historical VR worlds and interact with objects like books or artifacts.
- Educational missions trigger at specific locations, giving points, hints, or story rewards.
- System tracks completed missions, earned points, and individual progress.
- Activities align with the Grade 7–9 syllabus and maintain historical accuracy.



# Non-Functional Requirements

- App runs smoothly on Meta Quest 3 without crashes and supports offline use.
- Controls are simple and intuitive, allowing beginners to start quickly.
- VR experience avoids sudden camera movements; missions are short to reduce fatigue.
- Sinhala and English text/audio are clear, age-appropriate, and easy to understand.
- Content and code are maintainable, allowing updates and addition of new worlds or languages.





# Gantt Chart



# References



- [1] J. Brown, “Virtual Reality Utilisation in History Education: A Systematic Quantitative Literature Review,” International Journal of Learning and Teaching in Technology, vol. 11, no. 4, 2022. [Online]. Available: <https://journals.librarypublishing.arizona.edu/itlt/article/id/5995/>
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- [3] P. Holmes, “Past Lives, Present Learners: Future Directions for History Education in Virtual Reality,” ResearchGate Preprint, 2023. [Online]. Available: [https://www.researchgate.net/publication/393858141\\_Past\\_lives\\_present\\_learners\\_Future\\_directions\\_for\\_history\\_education\\_in\\_virtual\\_reality](https://www.researchgate.net/publication/393858141_Past_lives_present_learners_Future_directions_for_history_education_in_virtual_reality)
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# Learning Mathematics in Gamified VR Educational Platform With History

IT22561602

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Specialization: Interactive Media



# Proven Gap & Creative Solution



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Sri Lankan middle school students, particularly in rural areas, struggle with math due to rote learning and abstract teaching methods. Concepts like algebra and geometry feel disconnected from real life, leading to low motivation, weak understanding, and poor performance.

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A gamified VR learning module for Grades 7–9 that embeds math lessons in culturally relevant environments. It includes bilingual support (Sinhala/English), adaptive difficulty, and gamification elements like points and badges to boost engagement, improve understanding, and support students in under-resourced schools.



# Research Gap

| Research Name                                             | Current Development (What they researched / where they stopped)          | Future & Special Features (Our Module Contribution)                                                          |
|-----------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| VR for Maths in Rural Middle Schools (Quasi-Experimental) | Showed score gains with basic VR; limited pedagogy and language support. | Add Sinhala–English content, gamified problem-solving, cross-links to science/history, and offline delivery. |
| Two VR Headsets for Teaching High-School Mathematics      | Compared hardware; no broad content or classroom model.                  | Optimize for Meta Quest 3 & PC, provide Sinhala–English lessons, gamified exercises, and offline mode.       |
| VR Apps in Math Classes: Attitudes Study                  | High enthusiasm; focused on attitudes, not full implementation.          | Target engagement gaps with inclusive Sinhala–English missions, curriculum mapping, and offline access.      |
| Immersive VR for Geometry Learning                        | Motivation improved in geometry; narrow topic scope.                     | Expand to full syllabus, Sinhala–English support, cross-subject tasks, analytics, and offline use.           |
| VR-Based Math Motivation: Review Protocol                 | Identifies gaps; still at planning stage.                                | Deliver a working bilingual, gamified math module with curriculum alignment and offline capability.          |

# Shortcomings of Existing Systems and the Proposed Solution

| Feature                                     | Prodigy Math Game | Prisms VR | Math World VR | NeoTrie VR | ClassVR | Proposed Solution |
|---------------------------------------------|-------------------|-----------|---------------|------------|---------|-------------------|
| Syllabus Alignment to Sri Lankan Grades 7-9 | ✗                 | ✗         | ✗             | ✗          | ✗       | ✓                 |
| Immersive Problem-Solving Across Topics     | ✗                 | ✓         | ✗             | ✗          | ✗       | ✓                 |
| Cultural Integration (Sri Lankan Contexts)  | ✗                 | ✗         | ✗             | ✗          | ✗       | ✓                 |
| Bilingual Support (Sinhala + English)       | ✓                 | ✗         | ✓             | ✗          | ✗       | ✓                 |
| Gamification & Motivation Design            | ✓                 | ✗         | ✓             | ✗          | ✗       | ✓                 |
| Offline Functionality                       | ✓                 | ✗         | ✗             | ✗          | ✗       | ✓                 |
| VR Optimization (Meta Quest 3 / Headsets)   | ✗                 | ✓         | ✗             | ✓          | ✓       | ✓                 |



# Application of Specialization-based Knowledge

## Key Domains of Specialization-based Knowledge



- Experiential learning in mathematics.
- 3D spatial reasoning and visualization.
- Game-based problem solving.

## Specific Latest Technologies in that Domain



- Unity 3D with adaptive gamification logic.
- Oculus XR for immersive math quests.
- Blender assets for cultural math contexts (temples, trade ports).
- Offline functionality for rural schools.



## How to Evaluate and Prove Success



- Track pass rates and improvements in fractions, algebra, and geometry.
- Pre and post test comparisons in pilot schools.
- Engagement analytics such as retry rates and mission completion.
- Teacher reviews of syllabus alignment.

## Data Availability and Ethical Clearance

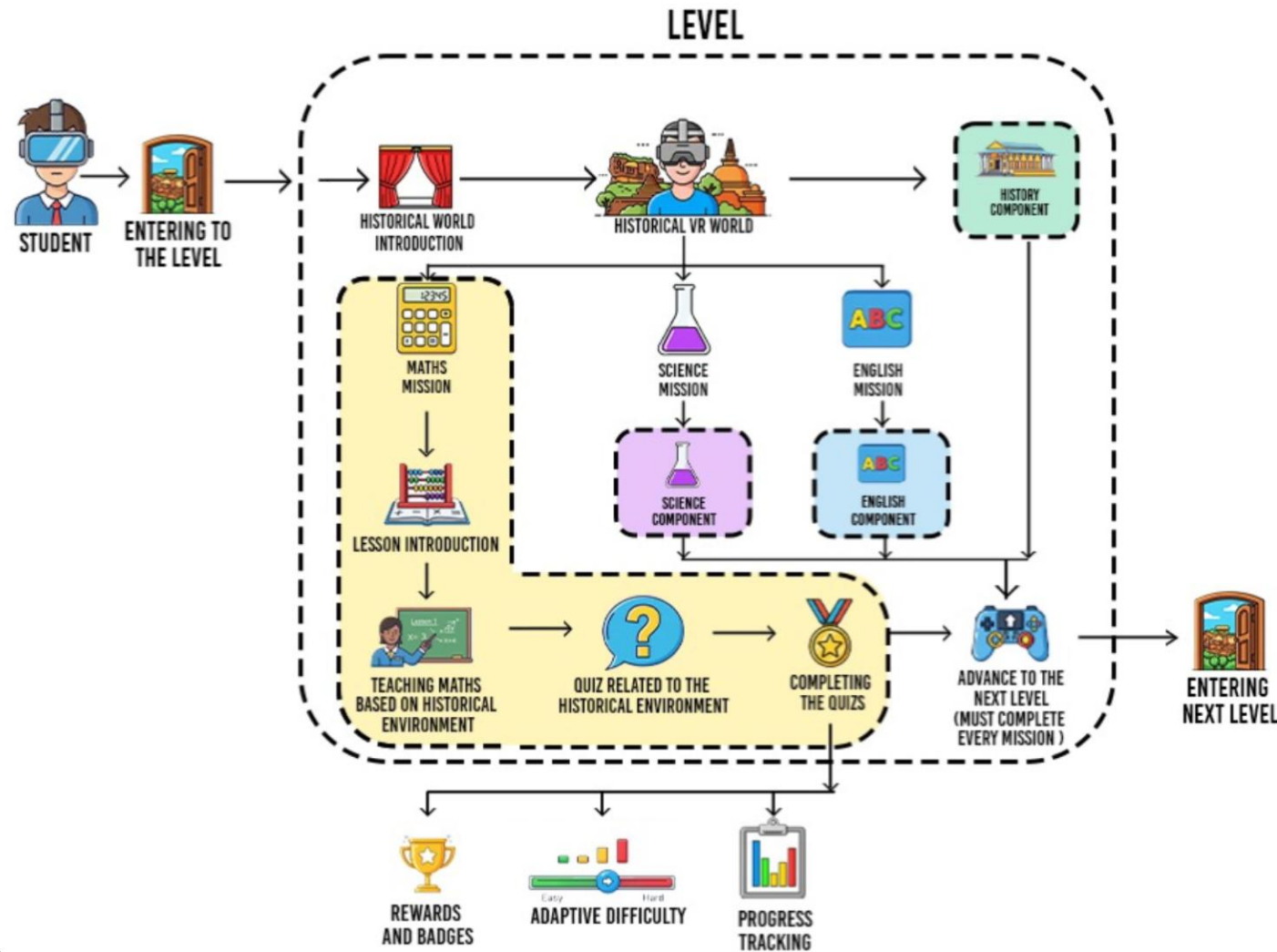


- Ministry of Education syllabus.
- Exam pass rate statistics showing high failure levels.
- Pilot testing with consented students and teachers.





# System Overview



# Real-world Scenario

- A Grade 9 student is placed in a virtual temple construction site.
- They solve geometry problems to align temple walls.
- Correct answers unlock progress in the mission.
- Students can measure their improvements in understanding and accuracy.

# Functional Requirements

- The module must align with the Sri Lankan Grade 7–9 math syllabus, covering fractions, algebra, geometry, and word problems.
- Students should solve tasks in immersive 3D historical and cultural environments such as irrigation systems, trade ports, and temple construction.
- It must support interactive problem-solving: combining fraction pieces, using virtual balance scales, and drawing geometric figures.
- Include gamification with points, badges, trophies, and unlockable levels to keep students motivated.
- Track student performance (scores, time, hints, attempts) and provide teachers with dashboards and gallery mode for reviewing results.
- Offer Sinhala–English bilingual support with narration, text, and quick language switching for better accessibility.

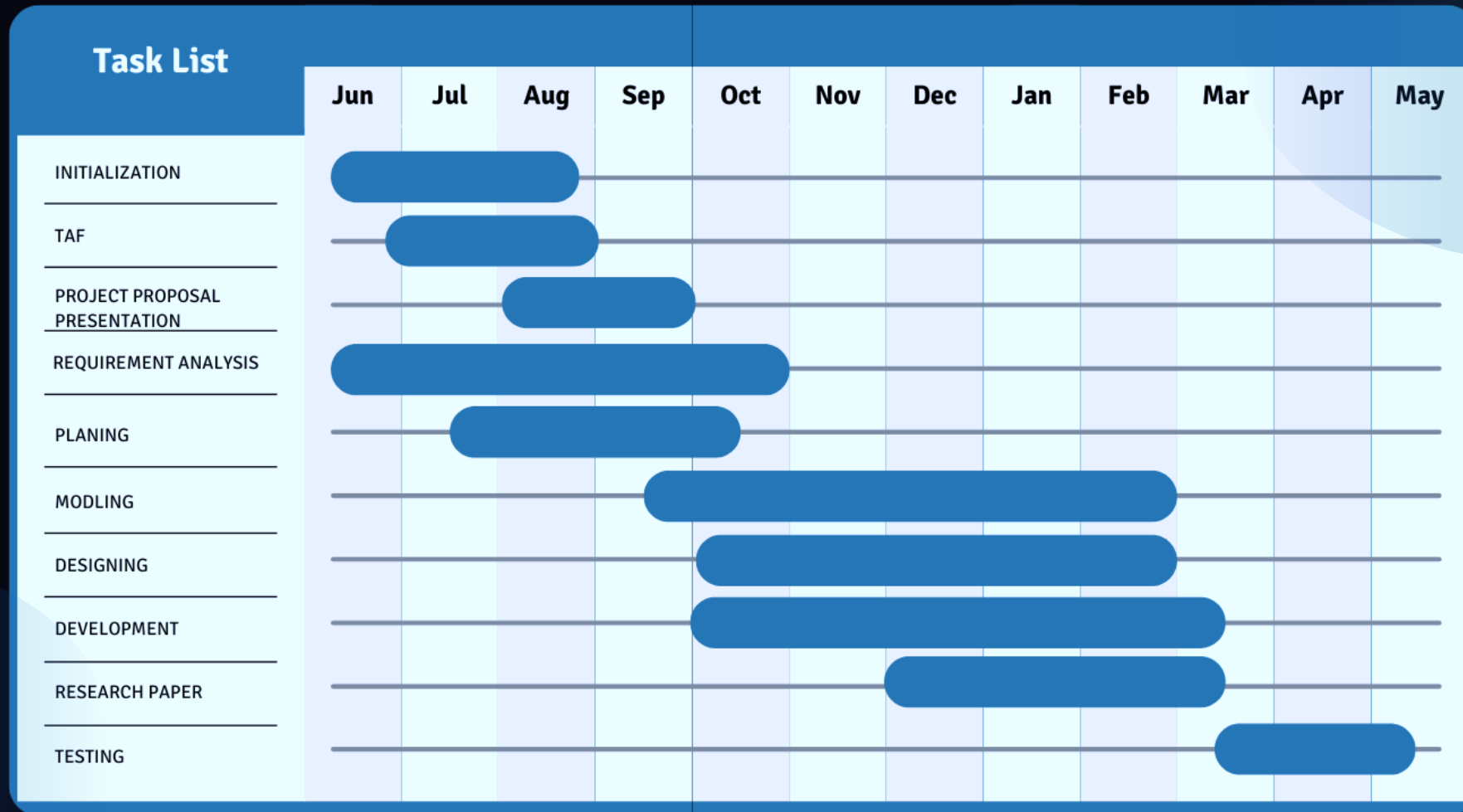


# Non-Functional Requirements

- The VR module must run reliably on Meta Quest 3 without crashes, with auto-save and checkpoint recovery.
- Scenes must load in under 5 seconds and maintain 72 FPS for smooth, safe VR use without motion sickness.
- The interface must be intuitive and child-friendly, with tutorials, large readable buttons, and culturally accurate visuals.
- The system must allow easy updates through modular design and editable JSON content for new topics or levels.
- Support multiple student profiles on one device and ensure compatibility with Quest 2 as well.
- Store all data locally, anonymize student identifiers, and protect privacy while working fully offline.



# Gantt Chart



# References



- [1] H. K. Wijesundara, “Assessing the Impact of Virtual Reality on Mathematics Teaching in Rural Middle Schools: A Quasi-Experimental Approach,” in Proc. 18th Int. Conf. Virtual Education, CEUR-WS, 2023. [Online]. Available: <https://ceur-ws.org/Vol-3691/paper17.pdf>
- [2] C. P. Vasile, “Exploring the Effectiveness of Two Types of Virtual Reality Headsets for Teaching High School Mathematics,” Eurasia Journal of Mathematics, Science and Technology Education, vol. 16, no. 8, 2020. [Online]. Available: <https://www.ejmste.com/article/exploring-the-effectiveness-of-two-types-of-virtual-reality-headsets-for-teaching-high-school-10996>
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# Learning Science in Gamified VR Educational Platform With History

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Specialization: Interactive Media



# Proven Gap & Creative Solution



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Sri Lankan middle school students struggle to grasp key science concepts due to limited hands-on learning, reliance on rote memorization, lack of lab facilities, and bilingual language challenges. Existing VR tools are not tailored to the Sri Lankan curriculum, lack bilingual support, and do not integrate multiple subjects or cultural context.



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Gamified Science VR module aligned with the Sri Lankan Grades 7–9 syllabus, embedded in a multi-subject VR platform. It offers immersive bilingual (Sinhala/English) virtual labs within culturally authentic historic settings, enhanced with gamification to boost engagement. Developed for Meta Quest 3 and tested locally, it aims to improve understanding, motivation, and learning outcomes in a scalable, culturally relevant way.

# Research Gap

| Research Name                                                                      | Current Development (What they researched / where they stopped)                   | Future & Special Features (Our Module Contribution)                                                                   |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Virtual Labs in Science Teaching: Sri Lankan Teachers' Perspectives and Challenges | Found VR labs boost engagement; focused on teacher views, not classroom roll-out. | Deliver offline VR labs with Sinhala–English support, gamification, cross-subject links, and full syllabus alignment. |
| Transforming Science Education with VR: Immersive Representations Model (IRM)      | Proposed an evaluation framework; no practical classroom implementation.          | Build hands-on VR lessons with Sinhala–English UI/voice, gamified paths, cross-curricular tasks, and offline use.     |
| Virtual Reality in Science Education: A Descriptive Review                         | Positive results mainly in higher ed; weak school-level curriculum integration.   | Provide school-level, syllabus-mapped VR science with Sinhala–English support, gamification, and offline access.      |
| Adding Immersive VR to a Science Lab Simulation (Presence↑, Learning↓)             | VR raised presence but reduced learning vs desktop; lacks pedagogy fixes.         | Improve outcomes via guided interactions, Sinhala–English explanations, analytics, and mission-based design.          |
| Bridging Gaps in Science Education with Virtual Labs                               | Noted internet/teacher-training barriers; theory only.                            | Ship offline VR labs + Sinhala–English teacher guides, gamified student tasks, and curriculum alignment.              |



# Shortcomings of Existing Systems and the Proposed Solution

| Feature                                                           | Veative (STEM VR) | VictoryXR Academy | Prisms VR (STEM) | Kai XR | Titans of Space PLUS | Proposed Solution |
|-------------------------------------------------------------------|-------------------|-------------------|------------------|--------|----------------------|-------------------|
| Syllabus alignment to Sri Lankan Grades 7–9                       | ✗                 | ✗                 | ✗                | ✗      | ✗                    | ✓                 |
| Integration of Math, Science, English tasks inside history scenes | ✗                 | ✗                 | ✗                | ✗      | ✗                    | ✓                 |
| Bilingual narration (Sinhala & English)                           | ✗                 | ✗                 | ✗                | ✗      | ✗                    | ✓                 |
| Scene length optimized for 5–10 min missions                      | ✓                 | ✓                 | ✓                | ✓      | ✓                    | ✓                 |
| Gamified progression with rewards                                 | ✓                 | ✓                 | ✓                | ✗      | ✗                    | ✓                 |
| VR optimized for Meta Quest 3                                     | ✓                 | ✓                 | ✓                | ✓      | ✓                    | ✓                 |
| Authentic Sri Lankan historical events                            | ✗                 | ✗                 | ✗                | ✗      | ✗                    | ✓                 |



# Application of Specialization-based Knowledge

## Key Domains of Specialization-based Knowledge



- Simulation and visualization in interactive media.
- Physics-based modelling of energy, matter, and ecosystems.
- Gamified instructional design.

## Specific Latest Technologies in that Domain



- Unity 3D with XR Interaction Toolkit and Oculus SDK.
- Physics engine for simple machines and energy flow.
- Blender models for lab equipment and natural systems.
- Bilingual narration system in Sinhala and English.





## How to Evaluate and Prove Success



- Pre and post tests on topics like photosynthesis and state of matters.
- Measure at least 20% improvement in test scores.
- Analytics on task completion, error rates, and time spent.
- Teacher evaluation of concept clarity.

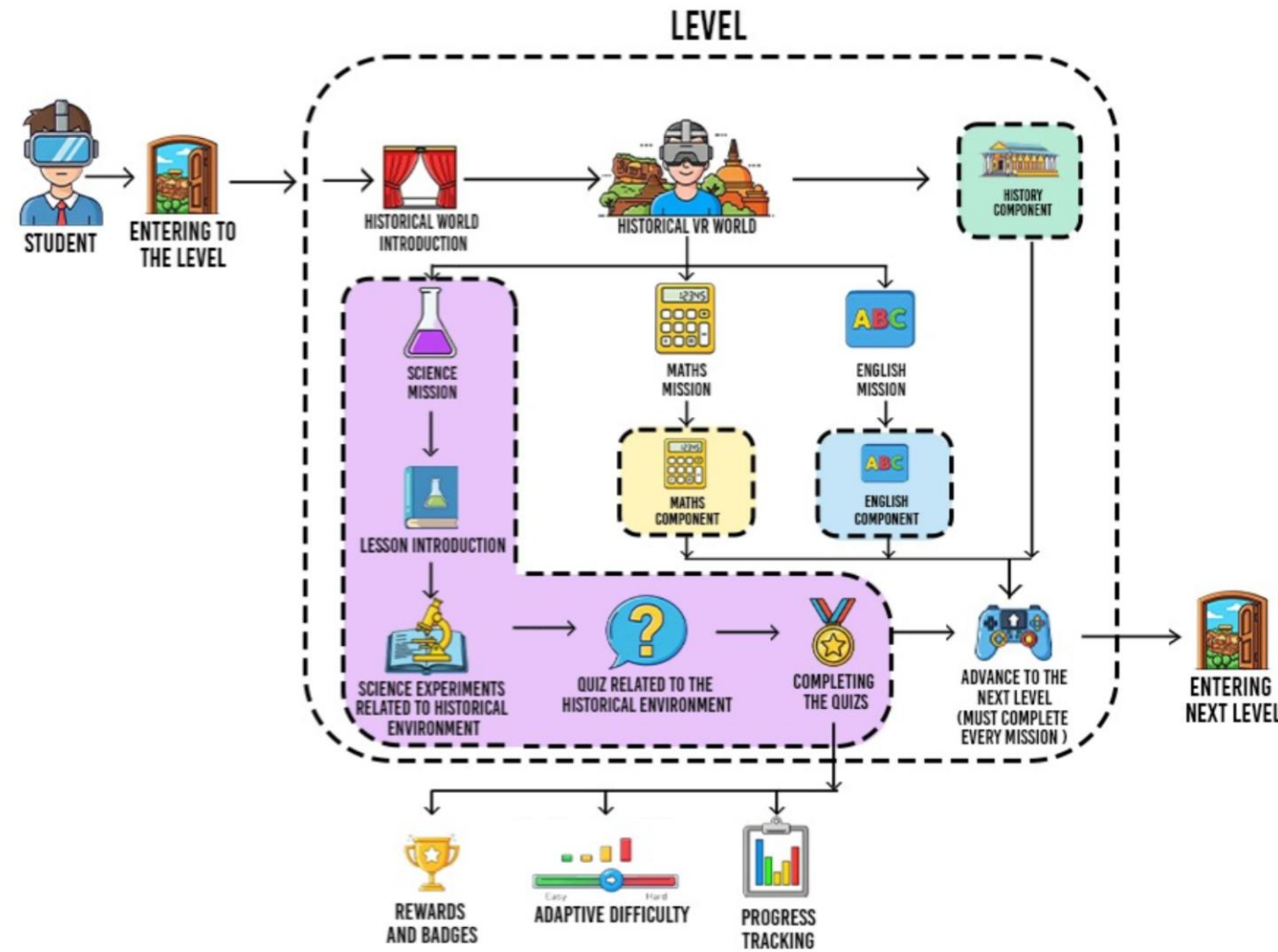
## Data Availability and Ethical Clearance



- National Grade 7–9 science syllabus and textbooks.
- Surveys from two pilot schools.
- Parental and school consent for student testing.



# System Overview



# Real-world Scenario

- A Grade 8 student enters a VR irrigation field during the Anuradhapura era.
- They perform a photosynthesis experiment with virtual plants.
- They receive immediate feedback and points for correct actions.
- Teachers review student results through progress data.

# Functional Requirements

- Virtual Experiments: Interactive VR labs for key science topics (ecosystems, machines, matter, plant biology).
- Guided Narration: Sinhala/English audio with step-by-step instructions and historical storytelling.
- Interaction & Feedback: Natural VR object use (grabbing, pouring, assembling) with instant feedback and summaries.
- Gamification & Tracking: Points, badges, levels, and data tracking for student progress and teacher evaluation.
- Bilingual Integration: Full Sinhala/English support across UI and narration, connected seamlessly to the multi-subject platform.



# Non-Functional Requirements

- Usability & Comfort: Child-friendly controls, stable 72+ FPS, short load times, and VR comfort practices.
- Safety: Guardian boundaries, headset warnings, and safe-modeled lab practices.
- Localization: Incorporate Sri Lankan culture, fauna, flora, attire, and narration style.
- Maintainability & Scalability: Modular Unity design for updates, scalable to multiple headsets and future grades.
- Compatibility & Evaluation: Optimized for Meta Quest 3 (portable to Quest 2/other XR), with activity logging for research and debugging.



# Gantt Chart





# References



- [1] S. Gamage, “Virtual Labs in Science Teaching: Sri Lankan Teachers' Perspectives and Challenges,” *Journal of Interactive Pedagogy and Education*, vol. 1, no. 2, 2018. [Online]. Available: <https://jipe.sljol.info/articles/10.4038/jipe.v1i2.13>
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- [5] R. Silva, “Bridging Gaps in Science Education with Virtual Labs,” *Journal of Interactive Pedagogy and Education*, 2021. [Online]. Available: <https://jipe.sljol.info/articles/13/files/6899945c5e608.pdf>

# Learning English in Gamified VR Educational Platform With History

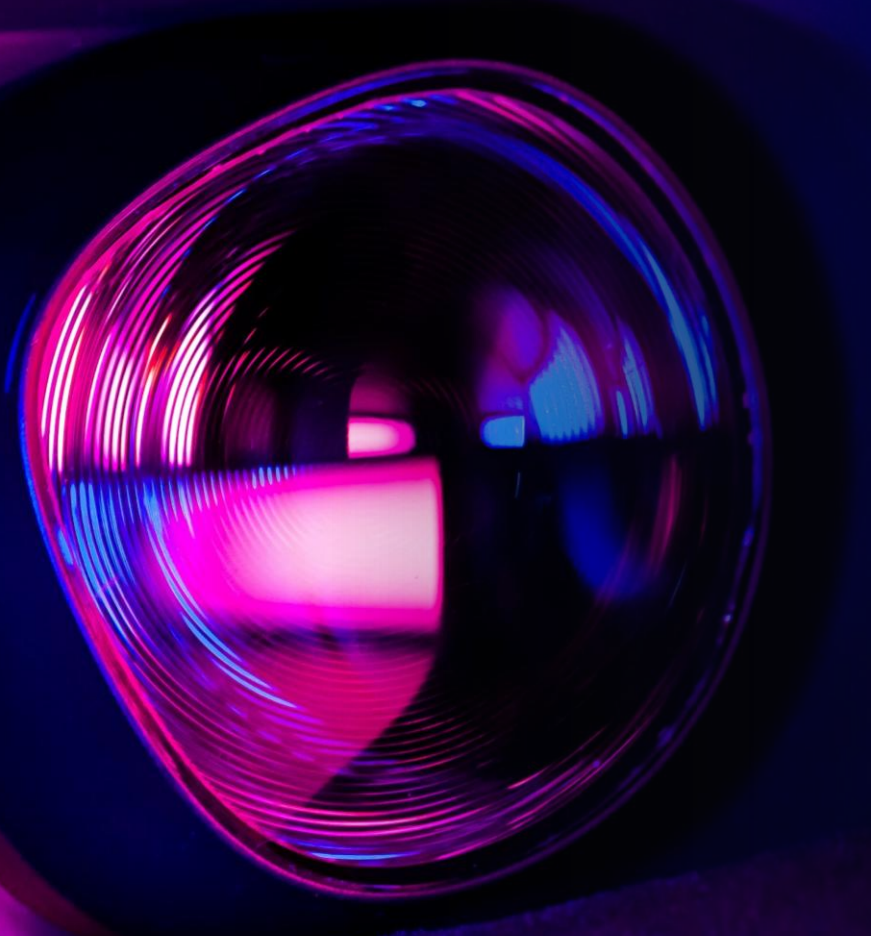
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Specialization: Interactive Media



# Proven Gap & Creative Solution



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Sri Lankan students in Grades 7–9 mainly study English for exams, focusing on grammar and reading with little real-life practice. Rural learners especially struggle with vocabulary, grammar, and confidence due to limited exposure outside school.

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A Gamified VR English Module will provide immersive, bilingual, and interactive tasks within historical contexts, making learning engaging and meaningful while boosting confidence, motivation, and practical language use.

# Research Gap

| Research Name                                               | Current Development (What they researched / where they stopped)   | Future & Special Features (Your Module Contribution)                                                            |
|-------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Gamification for Learning English in Sri Lanka              | Good potential noted; no VR or classroom deployment.              | Combine VR + gamification for English with Sinhala–English support, cross-subject scenes, and offline practice. |
| VR-Assisted Language Learning: Follow-Up Review (2018–2022) | Reviews benefits; limited diversity and local adaptation.         | Implement Sri Lanka–ready VR English with Sinhala–English UI/voice, curriculum mapping, and offline use.        |
| VR to Improve English Speaking: Student/Teacher Insights    | Enjoyable but modest gains; focused only on speaking.             | Cover all skills (listening/ reading/ writing/ speaking) with Sinhala–English scaffolds and mission tasks.      |
| Effects of VR-Assisted Language Learning (Meta-Analysis)    | Medium effects; ignores local language needs and practice models. | Design bilingual pedagogy, gamified progression, analytics, and syllabus-aligned tasks for Sri Lankan context.  |
| VR Game for Vocabulary Acquisition                          | Better motivation/learning for vocab only.                        | Extend beyond vocab to full English curriculum, Sinhala–English aids, cross-curricular content, and offline.    |



# Shortcomings of Existing Systems and the Proposed Solution

| Feature                                                                         | Mondly VR | ImmerseMe | Jayarathna et al. (Prototype) | ClassVR | Proposed VR English Module |
|---------------------------------------------------------------------------------|-----------|-----------|-------------------------------|---------|----------------------------|
| Syllabus Alignment (Sri Lankan Grades 7–9)                                      | ✗         | ✗         | ✗                             | ✗       | ✓                          |
| Cultural Relevance (Sri Lankan Contexts)                                        | ✗         | ✗         | ✗                             | ✗       | ✓                          |
| Bilingual Support (Sinhala + English)                                           | ✗         | ✗         | ✓                             | ✗       | ✓                          |
| Coverage of English Skills (Grammar, Reading, Comprehension, Speaking, Writing) | ✗         | ✗         | ✗                             | ✗       | ✓                          |
| Gamification & Motivation Design                                                | ✓         | ✓         | ✗                             | ✗       | ✓                          |
| Offline/Low-resource Accessibility                                              | ✗         | ✗         | ✗                             | ✗       | ✓                          |
| VR Optimization (Meta Quest 3 / Headsets)                                       | ✓         | ✓         | ✗                             | ✓       | ✓                          |





# Application of Specialization-based Knowledge

## Key Domains of Specialization-based Knowledge



- Language learning and bilingual instructional design.
- Gamification for engagement.
- Story-driven task design.

## Specific Latest Technologies in that Domain



- Unity 3D with interactive dialogue and text systems.
- Oculus Quest 3 for immersive environments.
- Sinhala-English narration and subtitle integration.
- Grammar and vocabulary engines inside story missions.



## How to Evaluate and Prove Success



- Vocabulary retention and grammar accuracy via pre and post tests.
- Comprehension and sentence building task performance.
- Student motivation and confidence surveys.
- Teacher feedback on practical classroom use.

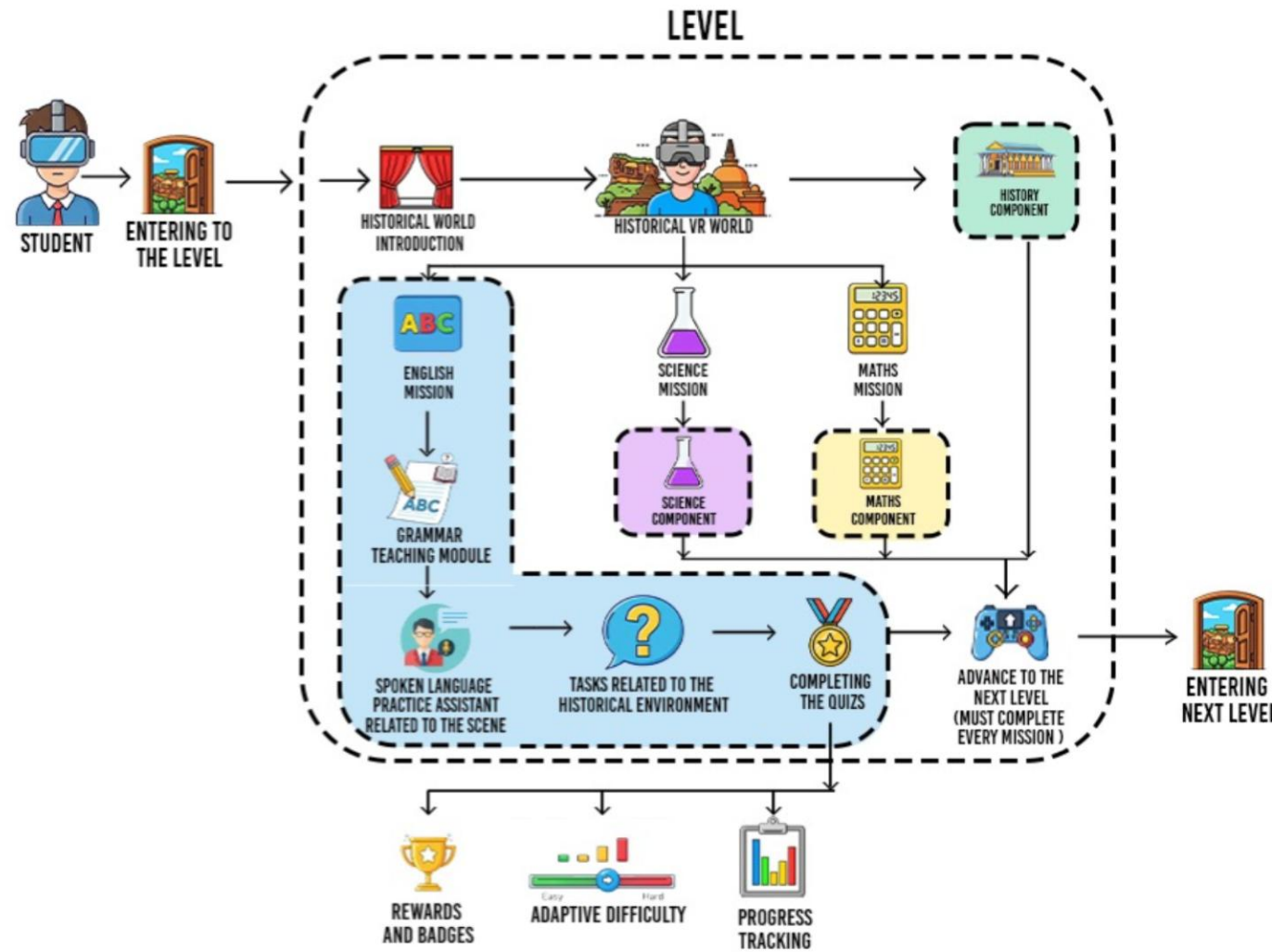
## Data Availability and Ethical Clearance



- Grade 7–9 English curriculum and teacher input.
- Surveys from Halwathura K.V. and Sri Dharmaloka K.V.
- Consent from schools and parents before pilot.



# System Overview



# Real-world Scenario

- A Grade 8 student enters a VR ancient marketplace.
- They role-play as a shopkeeper, using English sentences to interact with buyers.
- Points are awarded for correct sentence construction.
- Students can track their vocabulary and grammar improvements through the dashboard.

# Functional Requirements

- The module must follow the Grade 7–9 English syllabus with vocabulary, grammar, reading, and sentence formation tasks.
- Students should learn in 3D historical and cultural environments like Kingdoms and temple.
- It must support interactive tasks: object labeling, grammar correction, sentence re-ordering, and reading passages.
- Include gamification with points, badges, progress bars, and rewards to keep students motivated.
- Track individual progress and give teachers a dashboard to review results.



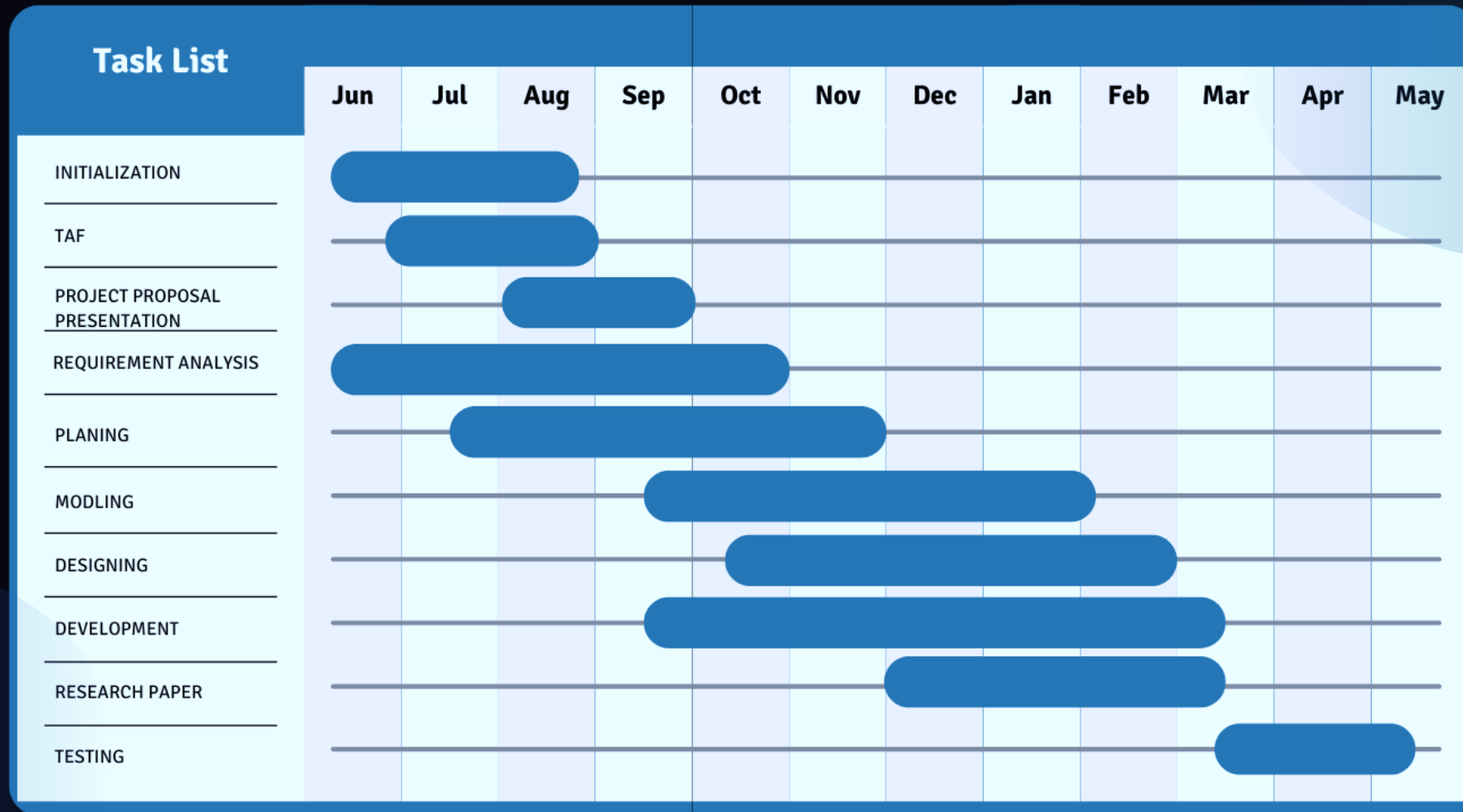
# Non-Functional Requirements

- The VR app must run reliably on Meta Quest 3 without crashes.
- Scenes must load in under 5 seconds and run at 60+ FPS smoothly.
- The interface must be simple and child-friendly, requiring little training.
- The system must allow easy updates for new syllabus content.
- Provide Sinhala–English narration and subtitles for better accessibility.





# Gantt Chart



# References



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# Commercialization & Market

## Ability of Commercialization

- The VR platform is syllabus aligned for Grades 7–9 in Sri Lanka.
- Bilingual support (Sinhala and English) makes it accessible for rural and urban schools.
- Scalable across public and private schools, tuition centers, and NGOs.
- VR headsets are already piloted in Sri Lanka, showing readiness for adoption.



## Investment / Cost and Recovery

| Item                                                     | Estimated Cost (LKR)              | Notes                                              | Recovery / Pricing Strategy                            |
|----------------------------------------------------------|-----------------------------------|----------------------------------------------------|--------------------------------------------------------|
| Development per module (Math, Science, English, History) | 5,000 – 10,000                    | Covers 3D assets, narration, testing, deployment   | Cost recovered through annual school subscriptions     |
| Total development (4 modules)                            | 20,000 – 40,000                   | Full platform build excluding hardware             | Breakeven within 2–3 years depending on adoption scale |
| Hardware (Meta Quest 3 headset)                          | ~180,000                          | Primary classroom device, one per school initially | Shared across subjects, one-time investment            |
| Other expenses (travel, teacher training, contingency)   | ~20,000 – 30,000                  | Pilot deployments and support                      | Covered through subscription margin or NGO funding     |
| Suggested pricing                                        | 10,000 – 15,000 per school / year | Affordable for schools, scalable with adoption     | Subscription and update fees ensure recurring revenue  |

## Target Market, Customer Profile, Value Proposition, IPR

| Category                     | Details                                                                                                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target Market                | Public and private middle schools in Sri Lanka (Grades 7–9). Tuition centers and NGOs focused on bilingual and STEM education.                                                                                    |
| Customer Profile             | School administrators looking for modern, syllabus-aligned solutions. Teachers needing practical VR modules. Students seeking interactive, bilingual learning.                                                    |
| Value Proposition            | Affordable VR platform aligned to national syllabus. Covers Math, Science, English, and History. Offers bilingual Sinhala-English support. Works offline for rural schools. Includes gamification for motivation. |
| Intellectual Property Rights | Ownership of custom 3D models, historical reconstructions, bilingual narration scripts, and VR gamification framework. Protectable via copyright and licensing for schools.                                       |





# Thank You!

Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka